

Benchmark Latency Report

Hardware: **NVIDIA RTX 5090** (Linux, native Triton) · Video: **combined.mp4** (4 541 frames) · Date: **2026-06-04** · Repo HEAD: 9172549b

All values are from real production benchmarks documented in docs/production_inference_benchmark.md. Only cycles with explicit **ACCEPTED** status are included (10 cycles). Behavior RTT values are available only for the 6 ensemble-era cycles (Cycle 9b B.2.b onward).

Per-Cycle Raw Data (accepted only)

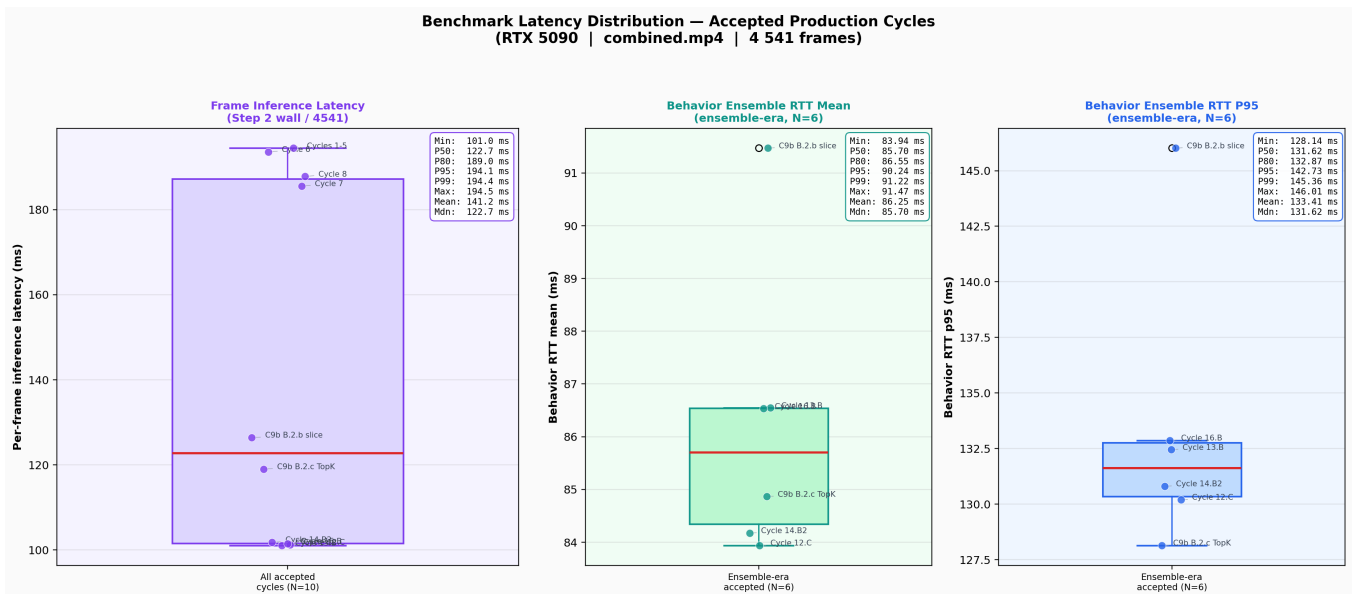
Cycle	Step 2 Wall (s)	Latency (ms/fr)	Step 2 FPS	Beh RTT mean (ms)	Beh RTT p95 (ms)
Cycles 1-5	883.000	194.451	5.143	—	—
Cycle 6	879.000	193.570	5.166	—	—
Cycle 7	842.600	185.554	5.389	—	—
Cycle 8	852.800	187.800	5.325	—	—
C9b B.2.b slice	573.927	126.388	7.912	91.470	146.015
C9b B.2.c TopK	540.399	119.004	8.403	84.865	128.138
Cycle 12.C	459.461	101.181	9.883	83.936	130.200
Cycle 13.B	458.696	101.012	9.900	86.545	132.448
Cycle 16.B	460.698	101.453	9.857	86.532	132.870
Cycle 14.B2	462.188	101.781	9.825	84.171	130.802

Summary Statistics (P50, P80, P95, P99, min, max, mean, median)

Statistic	Frame Latency (ms)	Equiv. FPS	Beh RTT Mean (ms)	Beh RTT P95 (ms)
Min	101.012	9.900	83.936	128.138
P50 (Median)	122.696	8.150	85.698	131.625
P80	188.954	5.292	86.545	132.870
P95	194.054	5.153	90.239	142.729
P99	194.371	5.145	91.224	145.358
Max	194.451	5.143	91.470	146.015
Mean	141.219	7.081	86.253	133.412
Median	122.696	8.150	85.698	131.625

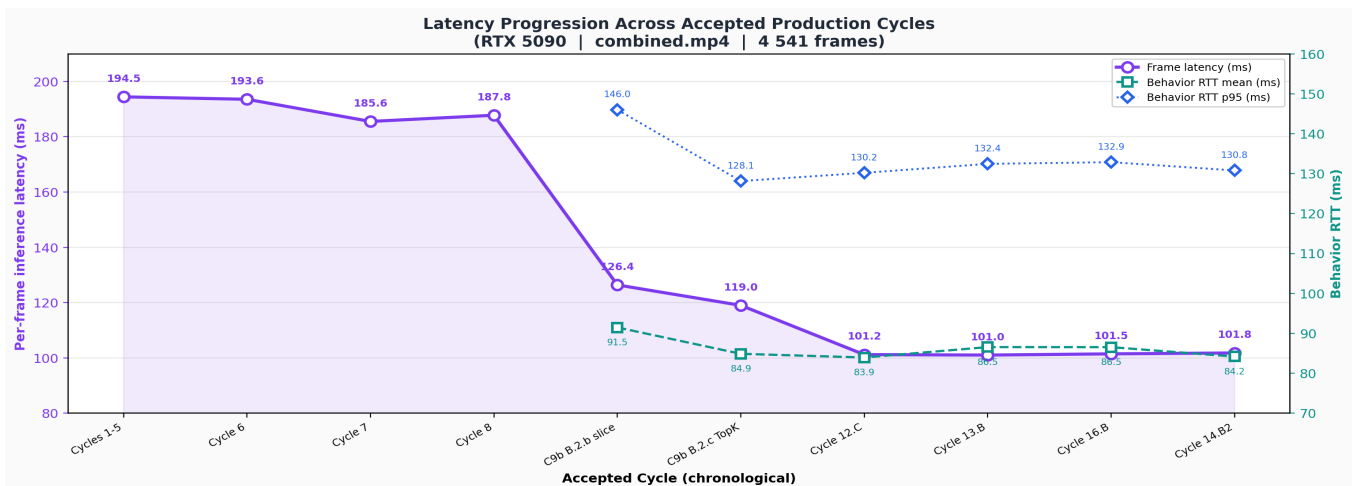
Box Plot — Latency Distribution

Left: per-frame inference latency across all 10 accepted cycles. Center: behavior ensemble RTT mean (6 ensemble-era cycles). Right: behavior ensemble RTT p95 (6 ensemble-era cycles). Red line = median; box = IQR; whiskers = full range; dots = individual cycles.



Latency Progression Across Accepted Cycles

Purple line (left axis): per-frame inference latency. Green squares (right axis): behavior RTT mean. Blue diamonds (right axis): behavior RTT p95. The sharp drop between Cycle 8 and C9b B.2.b marks the Triton ensemble migration (4 standalone gRPC calls → 1 ensemble call + server-side slice + Top-K adapters).



Frame Inference Latency — Detailed Statistics

Statistic	Value (ms/frame)	Interpretation
Min	101.0	Cycle 13.B (latest accepted stack)
P50 (Median)	122.7	Boundary between ensemble-era and pre-ensemble era
P80	189.0	Upper tail of pre-ensemble cluster
P95	194.1	Near Cycle 6 (pose chunking)
P99	194.4	Near Cycles 1-5 (initial accepted baseline)
Max	194.5	Cycles 1-5 (highest accepted latency)
Mean	141.2	Pulled up by the pre-ensemble cluster
Median	122.7	Same as P50

The distribution is **bimodal**: 4 pre-ensemble cycles cluster at 185–194 ms; 6 ensemble-era cycles cluster at 101–126 ms. The P50 (122.7 ms) sits at the transition point.

Behavior Ensemble RTT — Detailed Statistics

Statistic	RTT Mean (ms)	RTT P95 (ms)
Min	83.94	128.14
P50 (Median)	85.70	131.62
P80	86.55	132.87
P95	90.24	142.73
P99	91.22	145.36
Max	91.47	146.01
Mean	86.25	133.41
Median	85.70	131.62

Extremely tight clustering across all 6 ensemble-era accepted cycles: RTT mean ranges 83.9–91.5 ms; RTT p95 ranges 128.1–146.0 ms. The only outlier is C9b B.2.b (slice-only, before Top-K was added).

Pre-Ensemble Per-Model RTT (Cycles 1–5 Telemetry)

Model	Mean RTT (ms)
person_detector	9.58
rtmpose_model	41.08
posture_model	145.39
gaze_vertical_model	155.99
gaze_horizontal_model	168.54
gaze_depth_model	176.43

Before the Triton ensemble (Cycles 1–8), each behavior frame dispatched 4 separate gRPC calls. The slowest model (gaze_depth at 176 ms) bounded per-frame behavior wall. The ensemble collapsed these into a single call at ~85 ms.

Triton Server Compute (Cycle 14.D Probe)

Model / Ensemble	Server Avg (ms/exec)
gaze_depth_model	16.253
posture_model	16.352
gaze_vertical_model	16.389
gaze_horizontal_model	18.957
behavior_ensemble_gaze_slice_topk	29.976

Actual GPU computation per execution is 16–30 ms. The gap between server compute (~30 ms for the ensemble) and client-measured RTT (~85 ms) is gRPC transport, serialization, and queue wait.

Sub-Frame Micro-Latency Breakdown

Component	Latency	Source
Triton server compute (per exec)	11-15 ms	rtt_root_cause_investigation_77650001.md:80
Tensor construction (crop+lb+tobytes)	8.6 ms/frame	rtt_root_cause_investigation_77650001.md:192
True-batch concat (pre-memo)	6.6 ms/frame	rtt_root_cause_investigation_77650001.md:192
YOLO decode + NMS (all 68 crops)	1.82 ms/frame	rtt_root_cause_investigation_77650001.md:194
gc.collect + malloc_trim (pre-amort)	5-30 ms/frame	production_inference_benchmark.md:654
cv2 sequential read (no seek)	0.32 ms/frame	crop_frame_optimization_execution.md:350
cv2 seek-and-read (old pattern)	16.69 ms/frame	crop_frame_optimization_execution.md:351
LPM overhead (when enabled)	0.024 ms/frame	production_inference_benchmark.md:1113
Output deserialize (binary tensors)	~1 ms/frame	rtt_root_cause_investigation_77650001.md:213
gRPC transport+queue (4 boundaries)	~108-110 ms/fr	rtt_root_cause_investigation_77650001.md:186

Current Production State (Latest Accepted Profile)

Metric	Value	Source
Per-frame inference latency	~101 ms	Step 2 wall ~460 s / 4541 frames
Behavior ensemble RTT mean	84.2 ms	behavior_ensemble_gaze_slice_topk route
Behavior ensemble RTT P95	130.8 ms	Cycle 14.B2 accepted profile
Triton server compute	30.0 ms/exec	Cycle 14.D probe: ensemble average
Step 2 FPS	~9.8	4541 / 462.2 s
DB-completed FPS	~5.7	Cycle 14.B2: 4541 / 799.4 s
Avg GPU utilization	~12 %	Cycle 14.B2: 12.168 %
Peak GPU utilization	51–53 %	Cycle 14.B2 / 12.C / 16.B range

Remaining bottleneck: Step 2 through-pose wall is ~634 s (frame inference 462 s + pose tail 172 s). The pose tail is dominated by RTMPose provider batch wall (137 s). Further gains require either server-side compact postprocessing (Cycle 14.D) or video sharding (Cycle 15.B).

This PDF was generated by `scripts/build_latency_report_pdf.py`. Data source: `docs/production_inference_benchmark.md`. Re-run the script to regenerate after new cycles are accepted.